

Compact Lipschitz operators on unbounded spaces have only the periodic root spectrum

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Source Question

The source paper is Abbar, Coine, and Petitjean, *A note on the spectrum of Lipschitz operators and composition operators on Lipschitz spaces*, arXiv:2310.08965. Its first final open question asks whether the bounded-space spectral corollary remains true for unbounded metric spaces:

If $\widehat{f} : \mathcal{F}(M) \rightarrow \mathcal{F}(M)$ is compact, is the set A of periods of periodic points of f finite and

$$\sigma(\widehat{f}) \setminus \{0\} = \bigcup_{n \in A} \mathbb{U}_n?$$

Here, as in the source paper's Corollary 3.1 and Theorem 2.5 setup, A records periods of nonzero periodic points; the distinguished base point 0 does not yield a nonzero Dirac eigenvector.

Definitions and Source Facts

All metric spaces are pointed, complete, and have base point 0. If $f : M \rightarrow M$ is base-point-preserving and Lipschitz, $\widehat{f}$ denotes the induced bounded operator on the Lipschitz-free space $\mathcal{F}(M)$, defined on Dirac masses by $\widehat{f}(\delta(x)) = \delta(f(x))$. We write

$$\mathbb{U}_n = \{z \in \mathbb{C} : z^n = 1\}.$$

We use four facts from the source paper and from Abbar–Coine–Petitjean, *Compact and weakly compact Lipschitz operators*, arXiv:2110.03231.

1. If x is a periodic point of exact period n , then $\mathbb{U}_n \subset \sigma_p(\widehat{f})$.
2. If T is compact, then $\sigma(T) \setminus \{0\} = \sigma_p(T) \setminus \{0\}$, and 0 is the only possible accumulation point of $\sigma(T)$.
3. If $\widehat{f}$ is compact, then f satisfies the compactness characterization P_1, P_2, P_3 : images of bounded subsets are totally bounded, f is uniformly locally flat, and the tail condition P_3 holds. In particular, f is radially flat:

$$\frac{d(f(x), 0)}{d(x, 0)} \longrightarrow 0 \quad (d(x, 0) \rightarrow \infty).$$

4. The source paper uses cutoff operators $W_R = \theta_R \widehat{\text{Id}}$, where θ_R is 0 on $B(0, R)$, linear on the annulus $R \leq d(x, 0) \leq 2R$, and 1 outside $B(0, 2R)$. These satisfy $\sup_R \|W_R\| \leq 4$, $W_R \gamma$ is supported outside $B(0, R)$, and $(I - W_R) \gamma$ is supported in $B(0, 2R)$.

Proof Intuition

The source proof for flat-at-infinity maps cuts an alleged unbounded eigenvector into a far-out tail and tests the eigen-equation against a functional supported on that tail. Flatness at infinity makes the Lipschitz constant of $h \circ f$ on the tail arbitrarily small, forcing the nonzero eigenvalue to be arbitrarily small.

Compactness gives less than flatness at infinity, but it gives exactly enough. The P_3 tail condition says that escaping domain pairs with image separation comparable to domain separation must have image pairs with an accumulation point. Therefore, once at least one image remains far out, such a large quotient is impossible. This conditional tail-flatness is all the cutoff proof needs, because the testing function vanishes on the bounded part of the image. Nonzero eigenvectors are forced into bounded supports; after that, a large invariant closed ball reduces the problem to the bounded corollary already proved in the source paper.

Tail Estimate

Lemma 1 (escaping-image tail flatness). *Let $\hat{f} : \mathcal{F}(M) \rightarrow \mathcal{F}(M)$ be compact. For every $\varepsilon > 0$ there is $R_\varepsilon > 0$ such that whenever $x, y \in M$ satisfy*

$$d(x, 0) \geq R_\varepsilon, \quad d(y, 0) \geq R_\varepsilon, \quad \max\{d(f(x), 0), d(f(y), 0)\} \geq R_\varepsilon,$$

one has

$$d(f(x), f(y)) \leq \varepsilon d(x, y).$$

Proof. If not, there is $\varepsilon_0 > 0$ and, for each k , points x_k, y_k with $d(x_k, 0), d(y_k, 0) \geq k$, with at least one of $d(f(x_k), 0)$ and $d(f(y_k), 0)$ at least k , and with

$$d(f(x_k), f(y_k)) > \varepsilon_0 d(x_k, y_k).$$

The domain pairs escape to infinity. The image pairs cannot have an accumulation point in $M \times M$, since one image coordinate escapes to infinity. This contradicts condition P_3 in the compactness characterization of arXiv:2110.03231, which would force a subsequence with $d(f(x_k), f(y_k))/d(x_k, y_k) \rightarrow 0$. $\square$

Lemma 2. *If $\hat{f}$ is compact and $\hat{f}(\gamma) = \lambda\gamma$ for some $\gamma \neq 0$ and $\lambda \neq 0$, then $\text{supp}(\gamma)$ is bounded.*

Proof. Assume, toward a contradiction, that $\text{supp}(\gamma)$ is unbounded. Fix $\varepsilon > 0$. By radial flatness and Lemma 1, choose R so large that

$$d(f(x), 0) \leq \varepsilon d(x, 0) \quad (d(x, 0) \geq R),$$

$$f(B(0, 2R)) \subset B(0, R),$$

and the conclusion of Lemma 1 holds with $R_\varepsilon = R$.

Let $W_R = \theta_R \text{Id}$ be the source cutoff operator. Since $\text{supp}(\gamma)$ is unbounded, $W_R \gamma \neq 0$. Put

$$C_R = M \setminus B(0, R).$$

Choose $g_R \in \text{Lip}_0(\{0\} \cup C_R)$ with $\|g_R\|_L = 1$ and

$$\langle g_R, W_R \gamma \rangle = \|W_R \gamma\|.$$

Let $h_R = W_R^* g_R = \theta_R g_R$. Then $h_R \in \text{Lip}_0(M)$, $\|h_R\|_L \leq 4$, and h_R is supported outside $B(0, R)$.

The element $(I - W_R)\gamma$ is supported in $B(0, 2R)$, hence $\widehat{f}((I - W_R)\gamma)$ is supported in $B(0, R)$, where h_R vanishes. Therefore

$$|\lambda| \|W_R \gamma\| = |\langle h_R, \widehat{f}(\gamma) \rangle| = |\langle h_R, \widehat{f}(W_R \gamma) \rangle|.$$

It remains to estimate $h_R \circ f$ on $\{0\} \cup C_R$. If $x, y \in C_R$ and both $f(x), f(y) \in B(0, R)$, then $h_R(f(x)) = h_R(f(y)) = 0$. Otherwise at least one of $f(x), f(y)$ lies outside $B(0, R)$, and Lemma 1 gives

$$|h_R(f(x)) - h_R(f(y))| \leq 4 d(f(x), f(y)) \leq 4\varepsilon d(x, y).$$

For the pair $0, x$ with $x \in C_R$, either $h_R(f(x)) = 0$, or radial flatness gives

$$|h_R(f(x)) - h_R(f(0))| \leq 4 d(f(x), 0) \leq 4\varepsilon d(x, 0).$$

Thus

$$\|(h_R \circ f)|_{\{0\} \cup C_R}\|_L \leq 4\varepsilon.$$

Since $W_R \gamma \in \mathcal{F}(\{0\} \cup C_R)$, we get

$$|\lambda| \|W_R \gamma\| \leq 4\varepsilon \|W_R \gamma\|.$$

Dividing by $\|W_R \gamma\|$ gives $|\lambda| \leq 4\varepsilon$. As $\varepsilon > 0$ was arbitrary, $\lambda = 0$, a contradiction. $\square$

Main Result

Theorem 1. *Let M be a complete pointed metric space and let $f : M \rightarrow M$ be a base-point-preserving Lipschitz map. If $\widehat{f} : \mathcal{F}(M) \rightarrow \mathcal{F}(M)$ is compact and A is the set of all natural numbers n for which f has a nonzero periodic point of exact period n , then A is finite and*

$$\sigma(\widehat{f}) \setminus \{0\} = \bigcup_{n \in A} \mathbb{U}_n.$$

Equivalently, since $C_f = \widehat{f}^*$,

$$\sigma(C_f) \setminus \{0\} = \sigma(\widehat{f}) \setminus \{0\} = \bigcup_{n \in A} \mathbb{U}_n.$$

Proof. The inclusion $\supset$ is exactly the periodic-point eigenvector lemma from the source paper.

Conversely, let $\lambda \in \sigma(\widehat{f}) \setminus \{0\}$. Since $\widehat{f}$ is compact, $\lambda \in \sigma_p(\widehat{f})$, so there is $\gamma \neq 0$ such that $\widehat{f}(\gamma) = \lambda\gamma$. By Lemma 2, $\text{supp}(\gamma)$ is bounded.

Choose R so large that $\text{supp}(\gamma) \subset B(0, R)$ and

$$f(\overline{B}(0, R)) \subset \overline{B}(0, R).$$

This is possible by radial flatness. Let $M_R = \overline{B}(0, R)$, a complete bounded pointed metric space, and consider $f_R = f|_{M_R} : M_R \rightarrow M_R$. By the compactness characterization, $f(M_R)$ is totally bounded and f_R is uniformly locally flat. Hence the source paper's bounded/totally bounded spectral corollary applies to $\widehat{f}_R : \mathcal{F}(M_R) \rightarrow \mathcal{F}(M_R)$.

The element γ belongs canonically to $\mathcal{F}(M_R)$, and $\widehat{f_R}\gamma = \lambda\gamma$. Therefore the source corollary gives $\lambda \in \mathbb{U}_n$ for some exact period n of a periodic point of f_R , hence of f . This proves

$$\sigma(\widehat{f}) \setminus \{0\} \subset \bigcup_{n \in A} \mathbb{U}_n.$$

Finally, A must be finite. If infinitely many periods occurred, the periodic-point lemma would put the infinite set $\bigcup_{n \in A} \mathbb{U}_n$ into $\sigma_p(\widehat{f})$. This infinite subset of the unit circle has an accumulation point different from 0, contradicting compactness of $\widehat{f}$. The equality for C_f follows from $C_f = \widehat{f}^*$ and equality of spectra for an operator and its adjoint. $\square$

Verification and Novelty Notes

The proof uses only the source paper’s support/cutoff and periodic-point arguments plus the compactness characterization from arXiv:2110.03231. The new step is Lemma 1, which extracts from P_3 exactly the tail estimate needed by the source’s flat-at-infinity cutoff proof.

Cheap run indexes were searched for arXiv:2310.08965, the title keywords, “compact Lipschitz operator spectrum”, “roots of unity”, and related Lipschitz-free keywords; no duplicate packet was found. Web searches on 2026-06-29 for exact and close phrases around the final question found the source arXiv record but no later answer. Human review should check the cutoff support step in Lemma 2, especially the use of the source cutoff operator W_R and the passage to the bounded restriction.

References

- [1] A. Abbar, C. Coine, and C. Petitjean, *A note on the spectrum of Lipschitz operators and composition operators on Lipschitz spaces*, arXiv:2310.08965.
- [2] A. Abbar, C. Coine, and C. Petitjean, *Compact and weakly compact Lipschitz operators*, arXiv:2110.03231.